

Public Services Quarterly

ISSN: 1522-8959 (Print) 1522-9114 (Online) Journal homepage: www.tandfonline.com/journals/wpsq20

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To cite this article: Alehegn Adane Kinde, Assefa Chekole Addis & Getachew Gedamu Abebe (2021) Research data management practice in higher education institutions in Ethiopia, Public Services Quarterly, 17:4, 213-230, DOI: 10.1080/15228959.2021.1879707

To link to this article: https://doi.org/10.1080/15228959.2021.1879707

Published online: 06 Dec 2021.

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PUBLIC SERVICES QUARTERLY
2021, VOL. 17, NO. 4, 213–230
https://doi.org/10.1080/15228959.2021.1879707

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Research data management practice in higher education institutions in Ethiopia

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ABSTRACT

There is increasing pressure from funders, publishers, the public and other research organizations on researchers to improve their research data management. The main objective of the study was to examine the practices of research data management at higher education institutions in Ethiopia. The study was employed multi-stage sampling technique to select the actual sample unit of respondents. Eight first generation Universities and one Science and Technology University were selected based on purposive sampling technique. 390 faculty member and 119 stakeholder respondents were selected based on simple random and purposive sampling techniques. The findings revealed that majority of faculty members were used and produced spreadsheet and structured scientific and statistical research data formats. Majority of faculty members were used their personal device to hold their research data. The findings indicated that majority of faculty members were identified their needs of research data management services such as, data management template, storage facilities, research data management policy, guidelines and providing training. The level of awareness on research data management among faculty members was very low. The study concludes that proper research data management is not being practiced. Further research can be done on research data management framework at higher education institutions.

KEYWORDS

Data management plan; data management services; research data; research data management

Introduction

The number of published scholarly papers is gradually growing and there is a growing understanding of the importance, variety and nature of the data produced in research development. The increase of scholarly published papers contributed to the increasing demand for research data management (RDM) that was generated during the course of the research conducted (Redkina, 2019). There is increasing pressure from funders, publishers, the public, universities and other research organizations on researchers to improve their data management and sharing practices (Kennan & Markauskaite, 2015). Funding agencies have included Data Management Plan (DMP) submission criteria as part of a funding application (Kahn et al., 2014; Kiran, 2015; Sewerin et al., 2015). For example, in the United States the National Science Foundation (NSF) is asking for a one-to-two page DMP describing how researchers plan to work with their data. In

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Canada, submission of DMP as part of application for funding is becoming a common for faculty members. Funding agencies currently need a DMP that has data types generated or created, metadata standards, access and sharing polices, privacy protection requirements, confidentiality, security, intellectual property rights and archival and preservation plans (Sewerin et al., 2015). This is due to the fact that data (obtaining from experiments, interviews, observations, simulation, survey) is the foundation of scientific research and is imposed as a prerequisite by funders and journals in the context of data sharing (Amorim et al., 2017; Patel, 2016). Recent years have seen a significant rise in the value of RDM as an integral part good scientific practices as well as a part of funders' guidelines on open access and data sharing. RDM best practice mainly ensures the quality and reliability of research and maximizes its effect. In addition, the publication and re-usability of research data offers great advantages such as improving the reputation of researchers and institutions, meeting the criteria of funders and keeping with the open science agenda. It is the duty of researchers and their institutions to ensure that research data are well managed (Krahe et al., 2020; Morgan et al., 2017; Netscher & Katsanidou, 2018). This is good for industry, the public and researchers themselves, whose work will be much more widely recognized and appreciated(Huang, 2016; Wolski & Richardson, 2011).

RDM is defined as procedures that allow the curation, organization, integration and interoperability of data created during the scientific process or the course of a research cycle. It is a part of research process and helps to ensure that the data (raw or primary) is properly collected, organized, described, preserved and shared (Arias-Coello et al., 2018; Chigwada et al., 2017; Kurbanoğlu & Špiranec, 2013).

Research data refers to the any information that a researcher gathers whether raw, unprocessed, processed and calibrated, or reported output, during the process of studying a particular topic. Statistical results, experimental findings, and observations resulted from field experiments, surveys, interviews, and images are all examples of data from the research (Academy, 2020; Anilkumar, 2018; Morgan et al., 2017). Thus, RDM is present at all stages of the study, as it includes data creation, reuse planning, processing and organization, conservation, as well as protection of access, storage, recovery, sharing and reuse (Arias-Coello et al., 2018; Morgan et al., 2017). This reality has led researchers to have to acquire the capacity to manage their research data. Many funding agencies require public disclosure of the research data produced as part of a funded project. Many agencies have enforced data sharing requirements and formal DMPs. Likewise, many higher learning institutions need research data produced as part of the research project to encourage open access and quality of research (Arias-Coello et al., 2018; Wiorogórska et al., 2018). In different developed countries, RDM is defined by the Code for the Responsible Conduct of Research which recognizes that good RDM practice includes data ownership, data storage, data sharing and retention (Krahe et al., 2020; Ünal et al., 2019).

Several stakeholders are involved in the whole RDM process. These stakeholders include; librarians, researchers, IT professionals, legal experts, research administrators and academic unit (Amorim et al., 2017; Liu, 2013). An OCLC research report recommending universities wide RDM policy would include universities, research compliance offices, IT departments, researchers, academic units and university libraries (Erway &

OCLC Research, 2013). Such stakeholders play governing roles in the definition of data repository criteria for RDM. Especially stakeholders will have a major impact on how research data is being reused in the light of legal issues, data repository platform selection, data retention period (Liu, 2013; Wolski & Richardson, 2011).

For this study faculty or academic deans, library directorates, ICT directorates, research directorates and legal services directorates are selected for RDM stakeholders considering Ethiopian higher education institutions research process.

The main objective of this study was to examine the RDM practices of higher education institutions in Ethiopia and particularly it answered the followings research questions;

- Do the institutions have DMP and are researchers using DMP when conducting research?
- Where do researchers keep their data after a study is completed?
- What RDM support service institutions provide to faculty members?
- What is the level of awareness among faculty members about RDM?
- What are the perception and role of stakeholders on RDM services?

Materials and methods

A descriptive cross-sectional survey study was conducted among university faculty members and stakeholders to assess the current practices of RDM in higher education institutions in Ethiopia.

Description of the study area

Ethiopian higher education institutions

In Ethiopia, there are 45 public Higher Education Institutions at present to ensure academic, research and community services agendas. Ministry of Science and Higher Education of Ethiopia, formerly named Ministry of Education of Ethiopia categorized these institutions as first, second, third and fourth generation and Science and Technology Universities based on their time of establishment and centers of excellence. 9 of them were established over the past couple of decade of years, called first generation Universities. Recently, after this study was conducted, new classification also made among these institutions based on Ethiopian Education Development Roadmap form 2018 to 2030 years. These classifications are research Universities (8 in number also known as first generation Universities), Applied Science Universities (15 in number) and General Education Universities (20 in number) and Science and Technology Universities (2 in number).

Therefore, 8 first generation universities and Adama Science and Technology University were selected purposively using the criteria of being good research experience over 20 years and availability of ICT infrastructure that support RDM to attain the objectives of this study.

Accordingly, 9 of the following universities were selected located from Addis Ababa city administration and the four regional states of Amhara, Oromia, Tigray and SNNP are:

Addis Ababa University (AAU), Addis Ababa city administration; established 1951, University of Gondar (UoG), Amhara regional state; established 1954, Bahir Dar University (BDU), Amhara regional state; established 1951, Jimma University (JU), Oromia regional state; established 1951, Mekelle University (MU), Tigray regional state; established 1991, Haromaya University (HRU), Oromia regional state; established 1951, Hawassa University (HU), SNNP regional state; established 1976, Arba Minch University (AMU), SNNP regional state; established 1951, Adama Science and Technology University (ASTU), Oromia regional state; established 1993; from Science and Technology University category.

Target population, sample and sampling technique

The target population of this study comprised of faculty members and stakeholders among nine purposively selected Universities. Faculty members (instructors) who are conducting the actual researches work. Stakeholders are the main actors who are involving in RDM process. In this study, Research Directorate, Library Directorate, ICT Directorate, Law Service Directorate and Faculty Deans are the stakeholders with total of 119 in number.

The sampling frame included all faculty members in the study areas. Hence, there are 15,164 faculty members; the total data was obtained from Ministry of Science and Higher Education Office. Out of 15,164 faculty members, 390 faculty members of sampling unit have been selected using Yemane's sampling determination formula using simple random sampling technique through random numbers selection from one list which consists of 15,164 all faculty members as a representative of Higher Education Institutions of Ethiopia.

Therefore, this study was employed multi-stage sampling technique to select the actual sample unit of respondents who are participated in the study.

The first stage of sampling was used purposive sampling technique to select nine first generation Universities using the criteria of being good research experience over 20 years and availability of ICT infrastructure that support RDM to attain the objectives of this study.

The second stage of sampling was used simple random sampling technique to select 390 sample units for this study through random numbers selection from one list which consists of 15,164 all faculty members.

Lastly, purposive sampling technique was employed to select each sample units of faculty members from each of the respective first generation universities using the criteria of being good research experience.

Additionally, all 119 stakeholders were selected from 9 targeted Universities to achieve the objective of the study.

The following table illustrates that 390 sample unit's selection allocated to each selected universities using simple random sampling technique through random numbers from the total sampling frame 15,164 of faculty members of one list (Table 1).

Table 1. Summary of sampling plan.

No	Universities	Sampling frame	Sample units
1.	Addis Ababa University		75
2.	University of Gondar		75
3.	Adama Science and Technology University		16
4.	Bahir Dar University		39
5.	Hawassa University		36
6.	Jimma University		45
7.	Haramaya University		28
8.	Mekelle University		45
9.	Arbaminch University		31
	Total	15,164	390

Survey

Before the study was conducted, a five day workshop entitled "Research Data Stewardship and e-Infrastructure for Ethiopian Stakeholders" was conducted on 13 Aug 2018 at the University of Gondar. The focuses of the workshop was RDM and the investigators of this research were the participant of the workshop. Most of the survey of this research is taken from the workshop discussion. The researchers understood that separate survey is necessary for faculty members and stakeholders since faculty members do not aware of the process of RDM rather the actual of the research work.

Therefore, the researchers were designed well-structured questionnaires for both faculty members and stakeholders separately that has three sections. The first section gathers demographic information including institutions, faculty, educational level and research experiences. The second section explored nature of research data including file format and data storage. The last section explored different issues of RDM such as principles and RDM services to assess the awareness level of researchers. All questions for faculty members were closed ended. Mainly the questionnaire for stakeholders' focused on RDM process and management issues and some of the questions were open ended. The questionnaires were written in English language. The researchers have sent and distributed the questionnaires to 390 faculty members and 119 stakeholders in line with the respective Universities through email and face to face communication.

Results

Survey result among faculty members

390 questionnaires were distributed for faculty members across the respective Universities. Out of which 350 questionnaires were returned, 337 fully completed questionnaires ready for analysis and interpretation and the remaining 13 were discarded due to incompleteness. Therefore, the response rate of faculty members survey were 86% ($n = 337$).

Background information

As shown in Table 2, the number of faculty member respondents received by each University were AAU constitutes 19.2% ($n = 65$),UoG: 19.8% ($n = 67$),MU: 12.4% ($n = 42$), JU:12.1% ($n = 41$), BDU: 9.7% ($n = 33$),HU:9.4% ($n = 31$), AMU:7.6% ($n = 25$), HRU:6.9% ($n = 23$), and ASTU:2.9% ($n = 10$).

Table 2. Background information of faculty members.

Characteristics	(*N*) (% of responses)
Institutions	
AAU	65 (19.2)
UOG	67 (19.8)
BDU	33 (9.7)
JU	41 (12.1)
HRU	23 (6.9)
HU	31 (9.4)
ASTU	10 (2.9)
MU	42 (12.4)
AMU	25 (7.6)
Academic rank	
Graduate Assistant	27 (6.8)
Lecturer	155 (43.8)
Assistant Professor	138 (43.1)
Associate Professor	17 (6.3)
Professor	0 (0)
College	
Medicine and health science	46 (13.6)
Natural and computational science	61 (18.1)
Agriculture and environmental sciences	38 (11.3)
Social sciences and humanities	49 (14.5)
Business and economics	39 (11.6)
Veterinary sciences	14 (4.2)
Engineering sciences	33 (9.8)
Informatics	26 (7.7)
Law	15 (4.5)
Education	16 (4.7)

From the survey result indicated that, the majority of faculty member respondents 43.8% ($n = 155$) had lecturer academic rank, followed by 43.1% ($n = 138$) assistant professors academic rank, 6.8% ($n = 27$) graduate assistances academic rank and 6.3% ($n = 17$) associate professors academic rank as shown in Table 2.

The majority of faculty member respondents were belongs to natural and computational science discipline with 18.1% ($n = 61$), followed by social science and humanities discipline with 14.5% ($n = 49$) and medicine and health science with 13.6% ($n = 46$) as shown in Table 2. In addition to this, 11.6% ($n = 39$) of faculty member respondents were belongs to business and economics college, agriculture and environmental sciences college of faculty member respondents with 11.3% ($n = 38$), engineering sciences college of faculty member respondents with 9.8% ($n = 33$), informatics discipline of faculty member respondents with 7.7% ($n = 26$), education college of faculty member respondents with 4.7% ($n = 16$), law college of faculty member respondents with 4.5% ($n = 15$), and the remaining 4.2% ($n = 14$) of faculty member respondents were belongs to college of veterinary sciences.

Research experiences

The survey seeks to know the research experiences of the respondents' (faculty members'). Respondents (faculty members) were asked to mention their research experiences by the questions of how long have the respondents been involved in research work and asked to specify the number of research work (number of different full length research study/studies) in number.

Table 3. Respondents' (faculty members) research experiences.

Characteristics	Number (%)
Respondents research experiences	
<5 years	81 (24.2)
5–10 years	170 (50.4)
11–15 years	68 (20.1)
>=15 years	18 (5.3)
Never involved research	0 (0)
No of research done	
1–4 Research	71 (17.9)
5–10 Research	172 (43.3)
11–15 Research	46 (11.6)
>=15 Research	43 (10.8)
Nothing	5 (1.3)

Table 4. Types of data format used and produced in research.

Types of data format used	Responses *N* (Percent)	Percent of cases
Textual data	72 (11.4%)	21.4%
Spreadsheet	225 (35.6%)	66.8%
Image data	86 (13.6%)	25.5%
Audio/video	68 (10.8%)	20.2%
Database data	29 (4.6%)	8.6%
Statistical data	152 (24.1%)	45.1%
Total	632 (100.0%)	187.5%

As a result, as shown in Table 3, most of the respondents had 5–10 years research experiences with 50.4% ($n = 170$), followed by less than 5 years research experiences with 24.2% ($n = 81$), 11–15 years research experiences with 20.1% ($n = 68$) and 15 and above years research experiences with 5.3% ($n = 18$).

Out of 337 respondents (faculty member), 43.3% ($n = 172$) number of respondents reported as they were conducted 5–10 different full length research studies, followed by 17.9% ($n = 71$) number of respondents were conducted 1 to 4 different full length research studies, 11.6% ($n = 46$) number of respondents reported as they were conducted 11 to 15 different full length research studies, and the remaining 10.8% ($n = 43$) number of respondents were conducted more than 15 different full length research studies as shown in Table 3.

Types of data format used

Based on the survey, the respondents (faculty members) were asked to mention the types of data format used for their research and produced in the process. Accordingly, vast majority of respondents 35.6% ($n = 225$) of respondents were used spreadsheet data format, followed by 24% ($n = 152$) of respondents were used structured scientific and statistical data (SPSS, GIS, etc.), 13.6% ($n = 86$) image data (JPEG, GIF, TIFF, PNG, etc.), 11.4% ($n = 72$) of respondents were used textual data, 10.8% ($n = 68$) audio or video data (Mp3, AVI, MP4, FLV data) and 4.6% ($n = 29$) of respondents were used database data (data in Access, Oracle, MySQL, etc.) used and produced. This question had multiple responses options and the result is analyzed in SPSS software using multiple responses (Table 4).

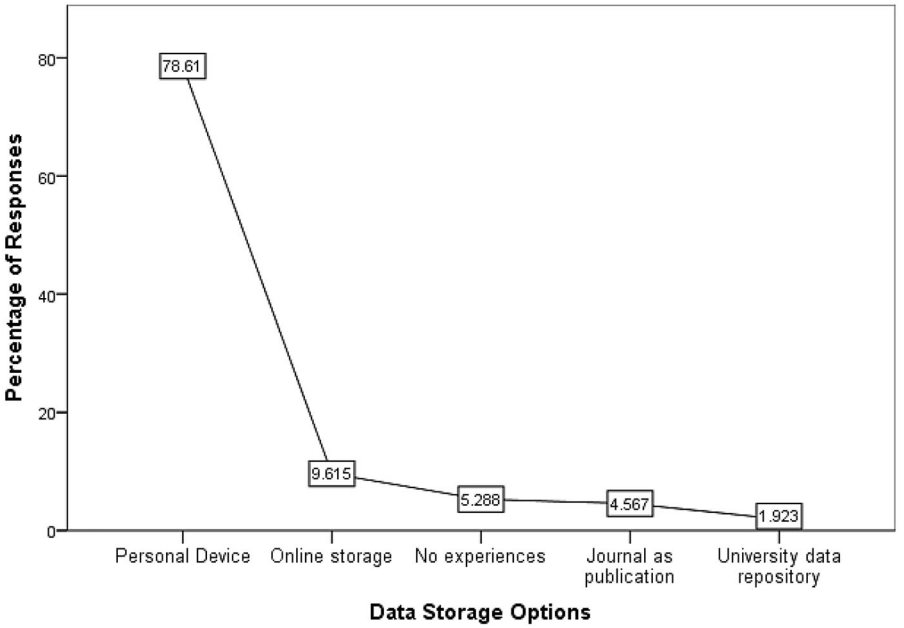


Figure 1. Research data storage.

Table 5. Knowledge about DMP.

	Frequency	Percent	Valid percent	Cumulative percent
Valid				
No	257	76.2	76.2	76.2
Yes	80	23.8	23.8	100.0
Total	337	100.0	100.0	

Data storage

The respondents (faculty members) were asked to select options where they store their research data upon completion of stud. Accordingly, the majority of the respondents, 78.6% ($n = 265$) were used personal devices (laptop, tablet etc.), followed by 9.6% ($n = 33$) of the respondents were used online storage (email, Google drive etc.), 4.5% ($n = 16$) of the respondents were used journal as publication (research data as a supplement file), 1.9% ($n = 6$)of the respondents were used University data repository. Out of this, 5.2% ($n = 17$) of the respondents reported as they didn't have experiences storing their research data after completion of the study. Journal as a publication mean uploading research data as a supplement file with the journal article on the publisher's website (Figure 1).

Practices of faculty members on DMP

The following Tables 5 and 6 respectively presents the current practices of the respondents' (faculty members) on knowledge and usage about DMP.

Table 6. Usage of DMP.

	Frequency	Percent	Valid percent	Cumulative percent
Valid				
No	322	95.5	95.5	95.5
Yes	15	4.5	4.5	100.0
Total	337	100.0	100.0	

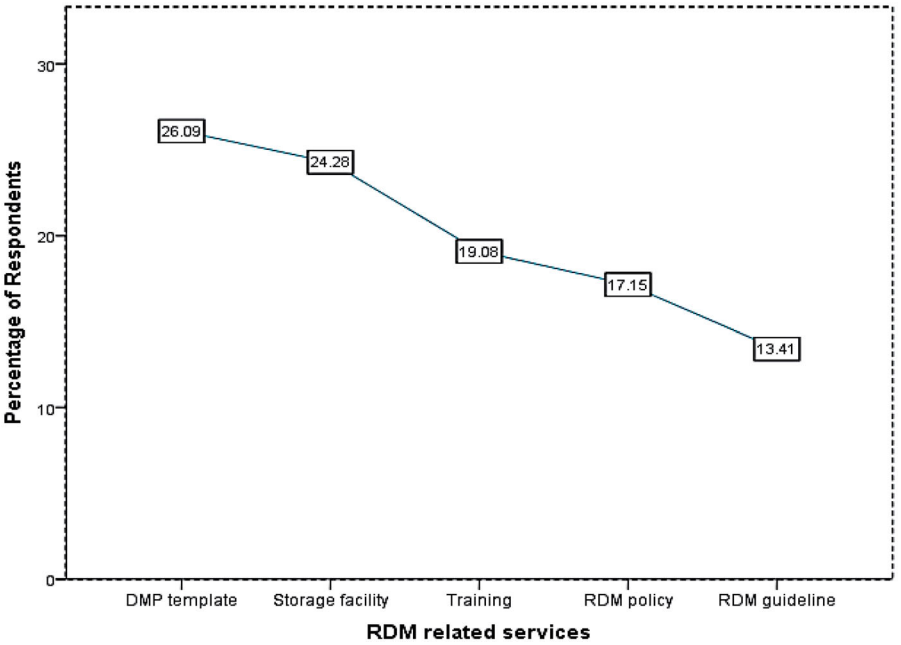


Figure 2. RDM services.

The survey result indicated that, 76.2% ($n = 257$) of respondents didn't know DMP and the remaining, 23.8% ($n = 80$) of respondents did know DMP.

The survey result indicated that, the majority of the respondents, 95.5% ($n = 322$) didn't used DMP for their research and the remaining 4.5% ($n = 15$) of respondents were used DMP for their research.

Needs of RDM services

Respondents (faculty members) were asked to select their needs among RDM services for the success of RDM at their institutions. As a result, 26% ($n = 216$) of respondents were needs DMP template, 24% ($n = 201$) of respondents were needs storage facility, 19% ($n = 158$) of respondents were wants to take RDM training, 17% ($n = 142$) of respondents were needs RDM policy and 13% ($n = 111$) of respondents were needs RDM guidelines. This question has had a multiple response and the respondents have selected more than one option. The researchers analyzed using multiple responses in SPSS software (Figure 2).

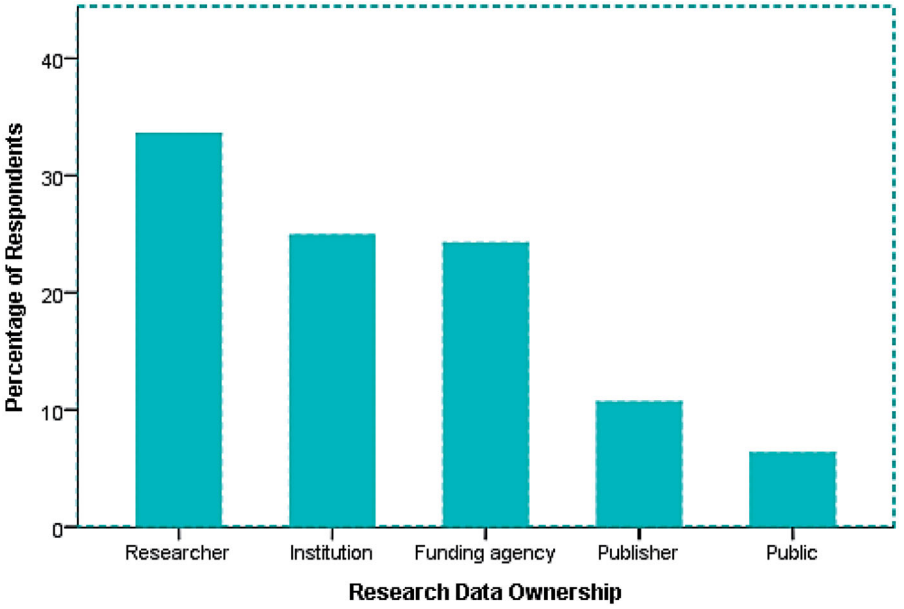


Figure 3. Opinion of respondents' (faculty members) on research data ownership.

Practices on research data ownership

The survey was asked respondents (faculty members) to give their opinion on the ownership of research data (Figure 3). Accordingly, as shown in Figure 4, 33.6% ($n = 248$) of the respondents were answered as researchers should own the data, 25% ($n = 184$) of the respondents were answered institutions should own the data, 24% ($n = 179$) of the respondents were answered funding agencies should own the data, 10.7% ($n = 79$) of the respondents were answered publishers should own the data and 6% ($n = 47$) of the respondents were answered the public should own the data.

Awareness on RDM

From the survey, the following statements were designed to measure the level of awareness of respondents (faculty members) about RDM shown in Table 7 below. The statements are consists of the major issues of RDM such as; RDM policy, DMP, metadata, research data sharing, research data quality, RDM ethical issues and principles.

From the survey result, the majority of the respondents 63.2% ($n = 213$) were replied disagree on the statement that refers knowledge on the existence of RDM policy in their institution. Similarly, the majority of the respondents 63.2% ($n = 213$) were replied disagree on the statement that refers knowledge on the concept of RDP. Regarding the statement that refers skill and knowledge to develop DMP, the majority of the respondents 69.4% ($n = 234$) were replied disagree. The survey result also indicated that, the majority of the respondents 77.2% ($n = 260$) were replied disagree on the statement that refers used a DMP for their research. Out of 330 respondents, 46.6% ($n = 157$) were replied disagree on the statement that refers had skill and knowledge about auditing the research data quality. On contrast, 44.8% ($n = 151$) of respondents were replied agree on the statement that refers familiar with the term metadata, and 43.0% ($n =$

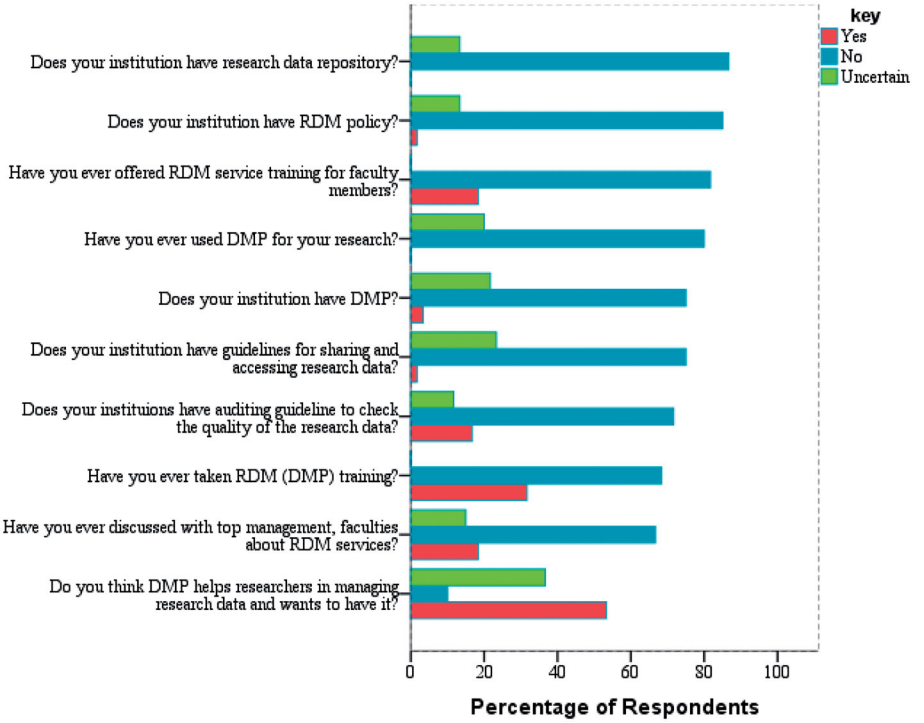


Figure 4. Opinions of stakeholders on RDM services.

145) were replied agree on the statement that refers skill and knowledge about auditing the research data quality. Regarding the statement that refers skill and knowledge about the benefits of research data sharing, the majority of the respondents 53.2% ($n = 180$) were replied disagree, and 54.0% ($n = 182$) of the respondents replied disagree on the statement that refers skill and knowledge about the principles of research data sharing and access. 45.1% ($n = 152$) of respondents were replied agree on the statement that refers skill and knowledge about the principles of intellectual property rights, and 63.5% ($n = 214$) were replied agree on the statement that refers skill and knowledge about RDM ethical issues. Lastly, regarding the statement that refers participated training on RDM, the majority of the respondents 62.0% ($n = 209$) were replied disagree.

Above all, the level of awareness among faculty members were found unaware except for some points such as metadata, data quality and intellectual property right which were aware. This indicates that the level of awareness on RDM among faculty members is very low.

Survey result among stakeholders

119 questionnaires were distributed for stakeholders across the respective Universities. Out of which 72 questionnaires were returned, 60 fully completed questionnaires ready for analysis and interpretation and the remaining 12 were rejected due to incompleteness. Therefore, the response rate of faculty members survey were 50.4% ($n = 60$).

Table 7. Level of awareness toward research data management.

Statements	Strongly disagree	Disagree	Neutral	Agree	Strongly agree	Median
I know the existence of RDM policy in my institution	30 (8.9%)	213 (63.2%)	53 (15.7%)	26 (7.7%)	15 (4.5%)	2
I know what DMP is	33 (9.8%)	213 (63.2%)	11 (3.3%)	67 (19.9%)	13 (3.9%)	2
I have the skill and knowledge to develop DMP	38 (11.3%)	234 (69.4%)	38 (11.3%)	19 (5.6%)	8 (2.4%)	2
I have used a DMP for my research	23 (6.8%)	260 (77.2%)	39 (11.6%)	10 (3.0%)	5 (1.5%)	2
I have the skill and knowledge about how to use data repository	58 (17.2%)	157 (46.6%)	39 (11.6%)	57 (16.9%)	26 (7.7%)	2
I am familiar with the term metadata	10 (3.0%)	65 (19.3%)	53 (15.7%)	151 (44.8%)	58 (17.2%)	4
I have the skill and knowledge about auditing the research data quality	6 (1.8%)	77 (22.8%)	51 (15.1%)	145 (43.0%)	58 (17.2%)	4
I have the skill and knowledge about the benefits of research data sharing	93 (28%)	180 (53.2%)	34 (9.9%)	24 (7.5%)	4 (1.4%)	2
I have the skill and knowledge about the principles of research data sharing and access	91 (27.0%)	182 (54.0%)	33 (9.8%)	26 (7.7%)	5 (1.5%)	2
I have the skill and knowledge about the principles of intellectual property rights	24 (7.1%)	94 (27.9%)	35 (10.4%)	152 (45.1%)	32 (9.5%)	4
I have the skill and knowledge about RDM ethical issues	30 (8.9%)	214 (63.5%)	34 (10.1%)	43 (12.8%)	16 (4.7%)	2
I have participated training on RDM	56 (16.6%)	209 (62.0%)	36 (10.7%)	23 (6.8%)	13 (3.9%)	2

Table 8. Frequency distribution among stakeholders.

	Frequency	Percentage
Stakeholders		
Faculty deans	35	57
Library directorate	8	13
Legal service directorate	5	8
Research Directorate	7	11
ICT directorate	5	8
Total	60	100

Table 9. Educational level.

	Frequency	Percent	Valid percent	Cumulative percent
Valid				
Master's Degree	28	46.7	46.7	46.7
PhD	32	53.3	53.3	100.0
Total	60	100.0	100.0	

Table 10. Academic rank.

	Frequency	Percent	Valid percent	Cumulative percent
Valid				
Assistant Professor	33	55.0	55.0	55.0
Associate Professor	8	13.3	13.3	68.3
Lecturer	19	31.7	31.7	100.0
Total	60	100.0	100.0	

Frequency distribution

From Table 8 shown as, the majority of stakeholders were faculty deans with 57% ($n = 35$), library directorate 13% ($n = 8$) and research directorate 11% ($n = 7$) response rate. As there is only one stakeholder from each institution (library, research, legal service and ICT), the response rate tends to be low except for the deans of the faculty since more than one faculty deans at one institution.

Educational level

From the survey result, 53.3% ($n = 32$) of stakeholders had Ph.D. and the remaining 46.7% ($n = 28$) of stakeholders had master's degree (Table 9).

Academic rank

According to the survey result, 55.0% ($n = 33$) of stakeholders had assistant professor academic rank, followed by 31.7% ($n = 19$) of stakeholders had lecturer academic rank and the remaining 13.3% ($n = 8$) of stakeholders had assistant professor academic rank as shown in Table 10.

Opinions of stakeholders on RDM services

Figure 5 presents the opinions of stakeholders on various aspects of RDM services. Since the selected stakeholders are the right person who knows the current research activity at their institutions, the Figure shows the actual RDM service situations of Ethiopian higher education institutions. A majority of respondents, 53% ($n = 32$), agree

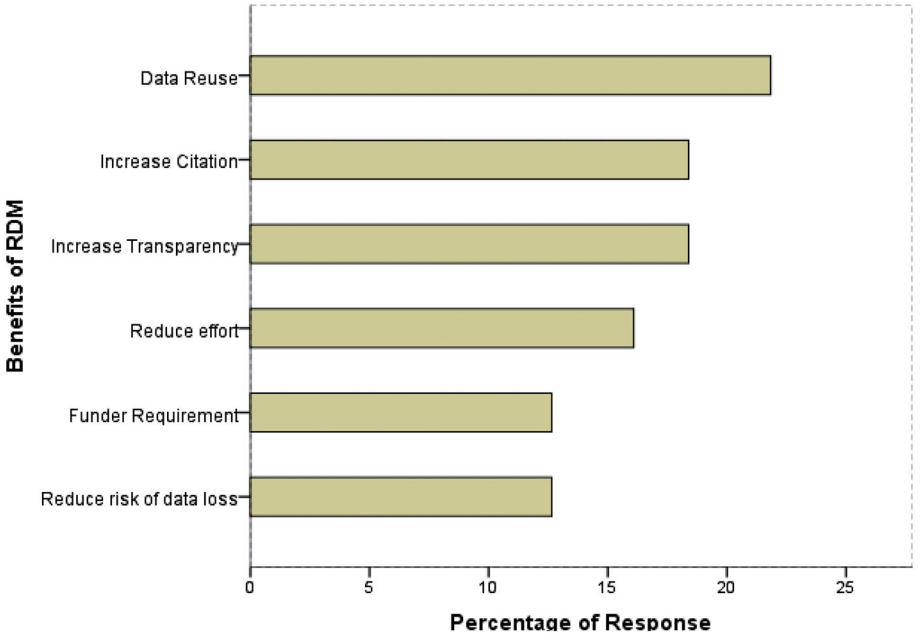


Figure 5. Opinion of stakeholders on benefits of RDM.

that as DMP helps researchers to manage research data but only 18% ($n = 11$) of respondents, given RDM related training for faculty members, 81 ($n = 49$), does not. The majority of the stakeholders, 66% ($n = 40$), have never discussed RDM with top research administrators, faculty members and colleagues, only 18% ($n = 11$) have discussed it. The majority of the stakeholders themselves, 68% ($n = 41$), do not take RDM-related training, only 31% ($n = 19$) did. As presented, it is possible to say that all selected institutions do not have RDM services available. For instance, in any of the institutions, there is no data repository. Almost all stakeholders are responded as they do not have DMP and do not use DMP when conducting research.

Opinions of stakeholders on benefits of RDM

Respondents were asked to give their opinion on the benefits of RDM for their institutions. Accordingly, as depicted in Figure 5, 21.8% ($n = 19$) answered important for data reuse, 18% ($n = 16$) helps to increase citation, 18% ($n = 16$) helps to increase transparency, 16% ($n = 14$), helps to reduce efforts, 12.6% ($n = 11$) to fulfill funder requirements and for reproducible.

Additionally, stakeholders were asked to briefly explain, using free text, about their role on RDM process at their institutions. Majority of the stakeholders stated as they do not have any specific role in RDM process including the librarians.

Discussion

The finding revealed that RDM remains a new concept for Ethiopia's higher education institutions. Researchers handle their own data, which was created from the study, using

their own personal devices. This contrasts with experiences from developed countries which reported that stakeholders particularly libraries took RDM as key components of their duties and responsibilities. Major results of the study are discussed below:

Research data format and RDM services

Most of the research data were in spreadsheet format, structured scientific and statistical data and image format. Additionally, audio/video, textual data and database formats also reported. This is important to consider various research data format during the development of data repository and DMP preparation. The UK data repository recognizes the availability of various research data format (Sewerin et al., 2015).

The respondents (faculty members) are confirmed that as they do not use DMP while they conduct a study. This is confirmed by stakeholders as Universities do not use and require DMP. On the contrary, reported in many countries, major funding agencies actually require applicants to submit a DMP as part of their study proposal (Anilkumar, 2018; Kahn et al., 2014; Kiran, 2015; Patel, 2016). A good DMP helps researchers outline what they are going to do with data during and after their study and it is also increasingly needed by funding agencies (Wiorogórska et al., 2018). The majority of respondents (faculty members) reported as they are using their personal device to hold their data after a study has been completed and some respondents do not have experiences to hold their data. This is because the institutions lack research data repository, central storage facilities and awareness in the institutions (Sanches, 2019). Legality concerns are one of the challenges for researchers to store their data in central storage (Netscher & Katsanidou, 2018).

Faculty members' mentioned, to start RDM, it is essential to have DMP template, storage facilities, training, RDM policy and guidelines. Stakeholders are also confirmed as they do not provide training for faculty members and do not yet have discussion about RDM with the institution's research management though they believe it is relevant. Previous studies identified that data storage, DMP, metadata description, policy, data repositories, data management training and support, and cultivating relationships were some of the recommended steps to get started of RDM (Henderson & Knott, 2015; Tripathi et al., 2017).Managing research data is not an easy task which needs designing RDM policy that clearly address copyright, data security, data privacy, data sharing, data preservation and data citation (Singh et al., 2018).

Data ownership

Regarding on research data ownership, respondents (faculty members) reported researchers, institutions, funding agencies, publishers and the public should own the data, in order. On the other hand, stakeholders reported as since data ownership will be varying based on the conditions, it has to be managed based on the institution RDM policy. Stakeholders also confirmed that as RDM policy does not available in the selected institutions. Section 2 of the Australian Code for the Responsible Conduct of Research states that "policies are required that address the ownership of research materials and data, their storage, their retention beyond the end of the project, and appropriate access to them by the research

community" (Kennan & Markauskaite, 2015). The National University of Singapore stated that the design and adoption of policies for RDM help to safeguard valuable data. The policy helps to answer allegations of research misconduct and assists in the protection of intellectual property (Kennan & Markauskaite, 2015).

A study conducted at the University of Ghana indicate the University of Ghana intellectual property policy defined data as a tangible research property and addresses the terms of data ownership as "Research data shall be jointly owned by the University and researcher(s) or determined on a case by case basis. Either party shall have a right to use the data for its research purposes … In spite of these provisions, entitlement to the ownership of primary data, software, and other products of research may vary, depending on the circumstances under which the research is conducted … " (Avuglah & Underwood, 2019).

Awareness of respondents on RDM

The majority of the respondents (faculty members) were unaware of the following RDM features such as how to develop DMP, benefits of RDM, usage of research data repository and principles of research data sharing which was crucial as stakeholders stat ed. The reasons may be lack of RDM training, lack of RDM policy, lack of data repository, lack of support and lack of professionals in each institution who took RDM as their role and responsibility. Faculty members and stakeholders should, therefore, be aware of RDM.

Role of stakeholders in RDM

Most of the stakeholders reported that they do not have any specific role in RDM process though they believe as DMP helps researchers to manage their data and RDM has many benefits. As different studies indicated that several stakeholders are involved in RDM process. These stakeholders include; librarians, researchers, IT professionals, legal experts, research administrators and academic unit. Some of the role of stakeholders are develop DMP template, develop RDM policy, develop research data repository, provide RDM training and support, prepare metadata, provide data storage (Amorim et al., 2017; Erway & OCLC Research., 2013; Liu, 2013).

Conclusion and recommendations

The finding of the study implies that proper management of research data is still lacking practices among faculty members and stakeholders of higher education institutions in Ethiopia. According to the results of this study, we noticed that lack of knowledge and skills to prepare and use of DMP, low level of awareness about data sharing, they handle their own data using their own personal devices, identifies RDM services among majority of faculty members. The majority of stakeholders confirmed that the need to develop and implement a RDM policy at national and institutional level. They were identified benefits of RDM and RDM services but their specific role in the RDM process was not well-known because of absence standardized RDM policy and framework of higher education institutions in Ethiopia. Stakeholders, in particular libraries need to work to raise awareness of proper institutional RDM. Authors recommended that, particularly, stakeholders to take

their part in RDM roles, the establishment of research data repository, preparation of DMP template, RDM policy, providing continuing training and support for faculty members. Further research can be done on RDM framework.

Acknowledgement

We would like acknowledge all respondents who filled out our questionnaires properly. We would like to also acknowledge the University of Gondar that supports this research by providing necessary expenses.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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